Chapter 18
The Greek Letters

Every chapter since Chapter 13 has rested on one idea: an option's price is the cost of manufacturing its payoff. This chapter verifies it by simulation: a delta-hedging program run 20,000 times costs, on average, the formula's value to within the simulation's own sampling error.

The same simulation produces a second number, and it is this chapter's subject: at five rebalances the standard deviation of the hedging cost is $1.21 on an option worth $3.19. The model assumes continuous, costless hedging; the market permits neither.

The Greeks are the vocabulary for managing that gap. They are not sensitivity analysis added afterward to a pricing model; Section 13.11 established that delta *is* the pricing calculation. This chapter takes the remaining derivatives of the same formula and shows that each answers a specific question a dealer must answer before the end of the day.

Section 18.1 states the dealer's problem. Sections 18.2 to 18.6 take the Greeks one at a time: delta, gamma, theta, vega, rho, and the rest. Sections 18.7 to 18.9 are the book: managing it, what desks actually hedge, and how often to rebalance. Sections 18.10 to 18.13 go past the Greeks to synthetic options, the feedback when hedging moves the market, and static hedging.

18.1 THE DEALER'S PROBLEM

Start where a dealer starts. A client buys 100,000 call options on a stock at $63, struck at $64, with eighteen weeks to expiration: $S_0 = 63$, $K = 64$, $T = 18/52$, $\sigma = 22\%$, $r = 4\%$, and no dividend. These are the inputs behind every number in this chapter. Chapter 15's formula values each option at $3.19438, so the dealer receives

$$100{,}000 \times 3.19438 = \$319{,}438$$

The dealer now has a liability that will be worth $\max(S_T - 64, 0) \times 100{,}000$ in eighteen weeks. What should they do?

Do nothing: a naked position. If the stock finishes at $54 the options expire worthless and the dealer keeps $319,438. If it finishes at $74 they owe $1 million, for a loss of $680,562.

Buy the stock: a covered position. Buying 100,000 shares at $63 costs $6.3 million. If the stock finishes at $74 the shares gain $1.1 million and the options cost $1 million, leaving a profit of $419,438. But if it finishes at $54, the shares have lost $900,000 and the dealer is down $580,562.

Stock at expiration	Naked	Covered
$54	+$319,438	−$580,562
$74	−$680,562	+$419,438

Figure 18.1 plots the two. *Neither is a hedge.* Each converts the dealer's position into a directional bet, merely in opposite directions, and each can lose roughly twice the premium collected. A dealer who intermediates thousands of such trades cannot run either.

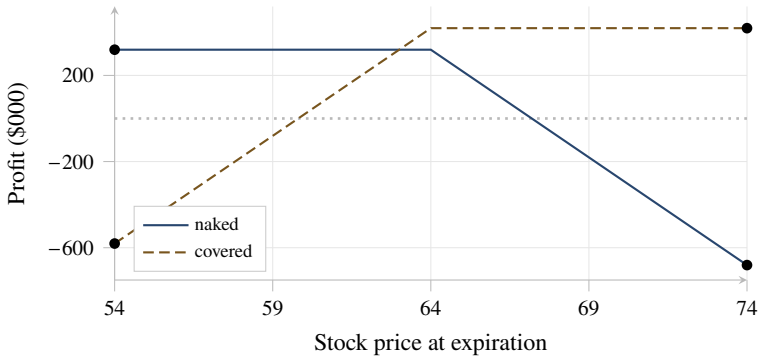


Figure 18.1: Profit at expiration on 100,000 written calls, $K = 64$, against the two unhedged responses. Marks are the table's two cases.

Switch between them: a stop-loss. The obvious repair is to run naked below \$64, buy the 100,000 shares when the stock rises through the strike, and sell them when it falls back. If every trade were done at exactly \$64, the dealer would hold the shares precisely when the option finishes in the money, the hedge would cost nothing, and the premium would be riskless profit. It cannot work. Purchases are made just above \$64 and sales just below, a stock near the strike crosses it again and again, and watching more closely makes each crossing cheaper but the crossings more numerous, so the hedge's error does not shrink. The stop-loss is the naive hedge that delta hedging improves on.

What is needed is a position that holds up in *both* rows of the table, whichever way the stock moves, and Chapter 13 already found it: hold a quantity of stock that makes the portfolio insensitive to the stock's movement. That quantity is delta.

Delta is the first of several quantities the dealer has to control. Over a short interval Δt the value Π of the position changes because the stock moves by ΔS, because time passes, and because implied volatility moves by $\Delta\sigma$. Expanding that change in the things that move gives

$$\Delta\Pi \approx \Delta\,\Delta S + \frac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t + \mathcal{V}\,\Delta\sigma \tag{18.1}$$

where delta Δ, gamma Γ, theta Θ (per year, with Δt in years) and vega $\mathcal{V}$ are the position's partial derivatives with respect to S, S twice, t and σ. A change Δr in the interest rate adds a term $\rho\,\Delta r$ in the same way, and for equity options it is usually small because the rate barely moves over the interval. Each Greek is the coefficient of one term, and hedging is the choice of offsetting positions that cancel terms of this expansion. Sections 18.2 to 18.6 take the terms in turn.

18.2 DELTA

Delta, the coefficient of the first term of (18.1), is the rate of change of an option's value f with respect to the price S of the underlying:

$$\Delta = \frac{\partial f}{\partial S} \tag{18.2}$$

For a European call on a non-dividend stock, Section 15.4 identified it already:

$$\Delta_{\text{call}} = N(d_1), \qquad \Delta_{\text{put}} = N(d_1) - 1 \tag{18.3}$$

so a call's delta lies in $(0, 1)$ and a put's in $(-1, 0)$. With a dividend yield, both are multiplied by e^{-qT}.

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In the dealer's problem, $N(d_1) = 0.5199$. Holding 51,995 shares against the 100,000 written calls makes the portfolio delta-neutral: a small move in the stock changes the value of the shares and the options by equal and opposite amounts. Example 18.1 reads delta as a multiplier and converts it into the option's leverage.

Every Greek in this chapter is evaluated at the option's own implied volatility, the convention Section 15.8 set for pricing, so a book's delta adds Black–Scholes deltas computed at different volatilities. That is exact only if each option's implied volatility stays put as the stock moves; Section 19.11 shows what changes when the smile moves with the spot, and equation (19.4) corrects the delta for it.

Example 18.1 Delta as a multiplier, and an option's leverage

Delta as a multiplier. Delta converts a move in the stock into a move in the option. If the stock rises by \$1 to \$64, the call should gain about $0.5199 \times 1 = \$0.52$; repricing at \$64 gives 3.7385, a rise of 0.5442. The gap, 0.0242, is curvature: the gamma term of (18.1), $0.04886/2 = 0.0244$ with the gamma of Section 18.3. For a \$3 fall delta predicts $-3 \times 0.5199 = -1.5597$, but repricing at \$60 gives 1.8568, a fall of only 1.3376: the value curve stays above delta's tangent.

Leverage. Dividing by the option's value turns the multiplier into percentages:

$$\frac{\Delta f}{f} \approx \Omega \frac{\Delta S}{S}, \qquad \Omega = \frac{S\,\Delta}{f}$$

where Ω, the option's **elasticity**, is its leverage: the percentage change in its value for a 1% change in the stock. Here $\Omega = 0.5199 \times 63/3.19438 = 10.25$: a 1% rise to \$63.63 raises the call by 10.56% exactly, and the call's volatility is about $\Omega\sigma \approx 226\%$. At the same stock price Ω is 5.71 at a strike of 54 and 15.98 at 74. Out-of-the-money calls carry the most leverage, which is why they are the speculator's instrument; Ω is also the $N(d_1)V/E$ that Section 24.6 computes for equity as a call on the firm.

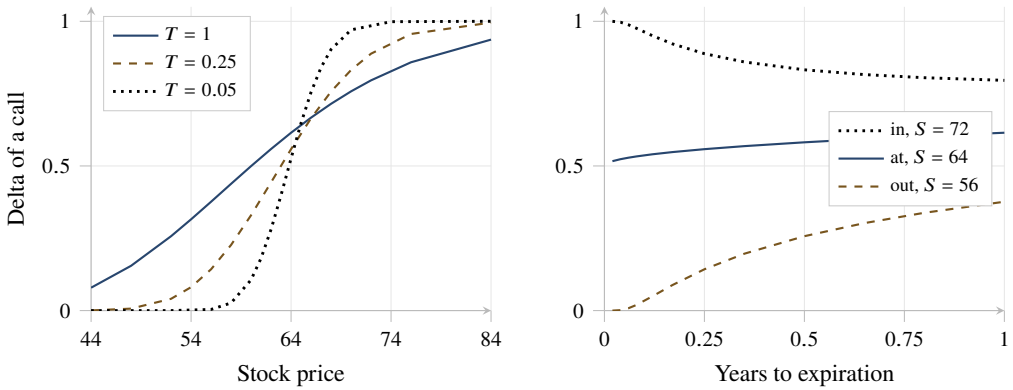


Figure 18.2: Call delta against the stock price and against remaining life. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Figure 18.2 shows the feature that makes hedging difficult. As expiration approaches, delta stops being a smooth function and approaches a step: an option that will finish in the money has delta 1, one that will finish out has delta 0, and near the strike the transition becomes almost vertical. A hedge tracking a near-vertical function must buy or sell almost the entire position on a small move in the stock. The right panel says it along the other axis: hold the stock still and delta walks to 1, to 0, or to neither as expiration nears. This is the pin risk of Example 10.6 and the 0DTE mechanism of Section 10.12, both now visible as statements about $N(d_1)$.

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A hedging program, week by week

The dealer's position is best understood by running it. Tables 18.1 and 18.2 follow the same written position (100,000 calls, $K = 64$, eighteen weeks) along two simulated price paths, rebalancing weekly to restore delta neutrality. The two paths differ in one respect only: the first finishes in the money and the second finishes out.

The bookkeeping is the same in both. The dealer borrows to buy $\Delta \times 100{,}000$ shares at week 0, and each week borrows or repays to bring the holding back to the new delta. The cumulative cost column compounds the previous week's balance at 4% and then adds the week's trade, so it is the dealer's outstanding debt.

Table 18.1: Weekly rebalancing against 100,000 written calls, path finishing in the money. $S_0 = 63$, $K = 64$, $r = 4\%$, $\sigma = 22\%$, 18 weeks. Costs in \$000, from unrounded prices.

Week	Stock price	Delta	Shares purchased	Cost of shares	Cumulative cost	Interest
0	63.00	0.520	51,995	3,275.7	3,275.7	2.5
1	61.12	0.421	−9,889	−604.4	2,673.8	2.1
2	62.51	0.488	6,651	415.8	3,091.6	2.4
3	60.86	0.394	−9,362	−569.7	2,524.2	1.9
4	58.46	0.261	−13,343	−780.0	1,746.1	1.3
5	61.57	0.418	15,788	972.1	2,719.5	2.1
6	62.02	0.438	1,915	118.8	2,840.4	2.2
7	62.03	0.431	−692	−43.0	2,799.6	2.2
8	61.55	0.391	−3,959	−243.7	2,558.1	2.0
9	63.48	0.513	12,183	773.4	3,333.5	2.6
10	61.94	0.396	−11,724	−726.2	2,609.8	2.0
11	64.41	0.574	17,822	1,147.9	3,759.8	2.9
12	64.71	0.597	2,360	152.7	3,915.4	3.0
13	63.86	0.523	−7,417	−473.7	3,444.7	2.6
14	63.25	0.455	−6,797	−429.9	3,017.5	2.3
15	66.43	0.781	32,561	2,163.0	5,182.9	4.0
16	70.36	0.988	20,695	1,456.1	6,642.9	5.1
17	69.67	0.998	975	67.9	6,716.0	5.2
18	66.04	1.000	238	15.7	6,736.8	
Shares delivered on exercise, proceeds $64 \times 100{,}000$					−6,400.0	
Cost of hedging the option					**336.8**	

Example 18.2 Reading the two hedging programs

The premium collected in Section 18.1 was \$319,438. Invested at 4% for eighteen weeks it would be worth \$323,900 at expiration, the number to compare the hedging cost against.

The hedge cost \$336,800 on the first path and \$309,300 on the second. One is 4.0% above that figure and the other 4.5% below. The dealer lost money on one path and made money on the other, having done nothing different in either case.

That is the whole content of the simulation below, seen transaction by transaction: discrete hedging does not reproduce the option payoff; it reproduces it *on average*, and the residual is the standard deviation Table 18.3 measures. Weekly rebalancing over eighteen weeks is eighteen adjustments, and eighteen is not many.

The trades themselves. In weeks 4 and 5 of Table 18.1 the stock fell to 58.46 and the dealer *sold* 13,343 shares, then rose to 61.57 and the dealer *bought* 15,788 back, \$3.11 higher, a loss of

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Table 18.2: The same position along a path finishing out of the money. Costs in $000, from unrounded prices.

Week	Stock price	Delta	Shares purchased	Cost of shares	Cumulative cost	Interest
0	63.00	0.520	51,995	3,275.7	3,275.7	2.5
1	61.60	0.446	−7,443	−458.5	2,819.7	2.2
2	63.45	0.536	9,080	576.1	3,398.0	2.6
3	61.36	0.421	−11,549	−708.6	2,692.0	2.1
4	61.57	0.426	474	29.2	2,723.2	2.1
5	61.08	0.390	−3,530	−215.6	2,509.7	1.9
6	59.75	0.305	−8,524	−509.3	2,002.3	1.5
7	58.01	0.201	−10,372	−601.7	1,402.2	1.1
8	58.38	0.205	346	20.2	1,423.5	1.1
9	61.22	0.358	15,323	938.1	2,362.6	1.8
10	58.88	0.197	−16,084	−947.1	1,417.4	1.1
11	62.68	0.440	24,278	1,521.8	2,940.3	2.3
12	64.54	0.584	14,383	928.3	3,870.8	3.0
13	64.24	0.558	−2,599	−167.0	3,706.9	2.9
14	66.55	0.765	20,684	1,376.5	5,086.2	3.9
15	64.57	0.594	−17,061	−1,101.6	3,988.5	3.1
16	63.73	0.484	−11,027	−702.8	3,288.8	2.5
17	62.86	0.292	−19,212	−1,207.7	2,083.7	1.6
18	60.90	0.000	−29,162	−1,776.0	309.3	
Option expires worthless; no shares held					0.0	
Cost of hedging the option					**309.3**	

about $41,500 on the shares just let go. Nothing was mismanaged: as Example 13.5 showed, delta hedging sells low and buys high by construction, and the accumulated cost of doing so is the option premium.

Why the paths end where they do. In Table 18.1 delta walks up to 1.000 and stays there, so the dealer ends holding the full 100,000 shares and delivers them on exercise. In Table 18.2 delta walks down to zero and the last trade liquidates the remaining position. The program converges to the terminal payoff automatically (no decision is made at expiration) because $N(d_1)$ becomes the step function of Figure 18.2.

Where the cost came from. On the out-of-the-money path the option expired worthless and the hedge still cost $309,300. In weeks 13 to 15 the stock went 64.24, 66.55, 64.57, ending within a third of a dollar of where it started, while the dealer bought 20,684 shares at 66.55 and sold 17,061 of them back at 64.57: about $33,800, a tenth of the premium, lost on one whipsaw. That is the gamma cost of Section 18.3, arriving before the concept is defined: a delta hedge pays for movement, not direction, and most for movement near the strike, where gamma peaks.

Portfolio delta. Delta is additive. A book of options on the same underlying has

$$\Delta_{\text{portfolio}} = \sum_i w_i \Delta_i$$

where w_i is the number of options of type i held, negative for those written. A desk holding thousands of positions therefore computes one number and trades one quantity of stock.

The price is the cost of the hedge

The two paths above showed the same program landing once above and once below the premium. Over many paths the claim behind every chapter since Chapter 13, verified on a two-step tree in

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Example 13.5, can be tested against the formula itself. Take the dealer's option of Section 18.1, worth \$3.1944, and run the program of Table 18.1 over 20,000 simulated paths, rebalancing n times and paying out $\max(S_T - K, 0)$ at expiration. The paths carry a *real-world* drift of $\mu = 10\%$, deliberately different from $r = 4\%$. Table 18.3 reports the cost in time-0 dollars.

Table 18.3: Cost of hedging 1 written call with no trading costs, in time-0 dollars. 20,000 paths; $\mu = 10\%$; formula value 3.1944.

Rebalances n	Mean cost	Std. dev.	Mean − formula
5	3.2079	1.2050	+0.0135
10	3.1946	0.8560	+0.0002
20	3.1950	0.6265	+0.0006
50	3.1972	0.4011	+0.0028
100	3.1942	0.2873	−0.0002
250	3.1949	0.1830	+0.0005
Formula (15.2)	3.1944	—	—

The mean cost is the formula's value. Once rebalancing is frequent enough the two agree to within a few tenths of a cent, which is the simulation's own sampling error. The dealer who charges \$3.19 and hedges diligently breaks even on average; that is what the number means, and why the formula is a price rather than a forecast. And the drift did not matter: the paths were simulated at 10% while the formula uses 4%, which is Chapter 13's principle verified numerically.

Discrete hedging leaves risk, and it falls as $1/\sqrt{n}$. At five rebalances the standard deviation is \$1.21, more than a third of the option's value; at 250 it is \$0.18. The hedge is exact only in the continuous limit, and the residual is what Chapter 15's assumption of continuous trading conceals. Kamal and Derman (1999) give its standard deviation as approximately $\sqrt{\pi/4}\,\sigma\mathcal{V}/\sqrt{n}$, with $\mathcal{V}$ the option's vega (14.77 here; Section 18.5). That is 0.407 at $n = 50$ and 0.288 at $n = 100$, within 2% of the table from 50 rebalances on; at five, where the approximation is coarsest, it overstates by 7%. That residual is where Section 18.9 begins: the model says hedge continuously, the market charges for every trade, and Table 18.7 adds the charge to exactly this simulation.

18.3 GAMMA

Delta hedging is exact only for infinitesimal moves, because delta itself changes. **Gamma** measures how fast:

$$\Gamma = \frac{\partial^2 f}{\partial S^2} = \frac{N'(d_1)}{S_0\sigma\sqrt{T}} \tag{18.4}$$

where N' is the standard normal density, the slope of N:

$$N'(x) = \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}.$$

Gamma is positive for both purchased calls and purchased puts, and negative for written ones.

The practical meaning is direct. A delta-neutral portfolio has no first-order exposure, so its profit and loss $\Delta\Pi$ over a move ΔS, to second order, is what remains of (18.1) once the delta term is hedged away and time and volatility are held still:

$$\Delta\Pi \approx \frac{1}{2}\Gamma\,(\Delta S)^2 \tag{18.5}$$

Figure 18.3 is the same statement geometrically. The delta approximation is the tangent

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at the hedge point; the option's value curves away from it on both sides, so a written option hedged with stock loses whichever way the stock goes.

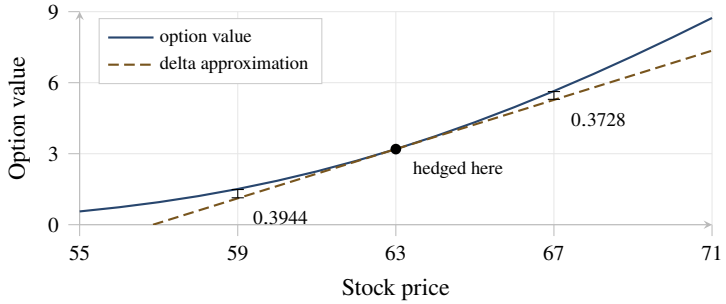


Figure 18.3: The dealer's option against its delta approximation at \$63. The marked gaps are Example 18.3's losses.

Example 18.3 What gamma costs a short option position

The dealer of Section 18.1 is delta-neutral at $S = 63$, with $\Gamma = 0.04886$ per option, so the book's gamma is $-0.04886 \times 100{,}000$, negative because the options are written.

Suppose the stock jumps before the hedge can be adjusted:

Stock	Actual P&L	$-\frac{1}{2}\Gamma(\Delta S)^2$
59	−0.3944	−0.3909
61	−0.0987	−0.0977
63	0	0
65	−0.0959	−0.0977
67	−0.3728	−0.3909

per option. The approximation is accurate to within two cents, and on 100,000 options a move to \$59 costs \$39,443 and a move to \$67 costs \$37,283.

Both columns are negative. A delta-neutral short-option position loses money whichever way the stock moves, and loses in proportion to the *square* of the move. Being short gamma means being short large moves regardless of direction, the same statement as Section 12.4's observation that a straddle buyer is betting on magnitude, seen from the other side of the trade.

What gamma is worth. Equation (18.5) puts a value on movement, and it depends on two things. The *sign* decides whether movement pays or costs: a book long gamma gains whichever way the stock goes, a book short gamma loses either way. The *size* decides how much, and because the relation is quadratic, doubling the move quadruples the amount. Section 18.4 sets that against what holding the position costs.

Figure 18.4 shows where the trouble concentrates. Gamma peaks at the money and rises sharply as expiration nears: at one year the peak is 0.031, at three months 0.058, at under three weeks 0.127, four times the one-year peak. At the money the contrast is sharper: with a week to run gamma is 0.20 against 0.027 for a one-year option (Figure 18.5), more than seven times. Gamma is the rate at which delta moves, so on the dealer's 100,000 options that means rehedging some 20,000 shares for every dollar the stock travels, against 2,700 for the one-year option, which explains most of what Section 10.12 said about same-day options.

Figure 18.5 shows the other dimension. At the money gamma rises without limit as expiration approaches, while away from the money it collapses toward zero. An option that will finish decisively in or out of the money has nothing left to hedge; one that will finish near the strike has all of it left, and no time in which to do it.

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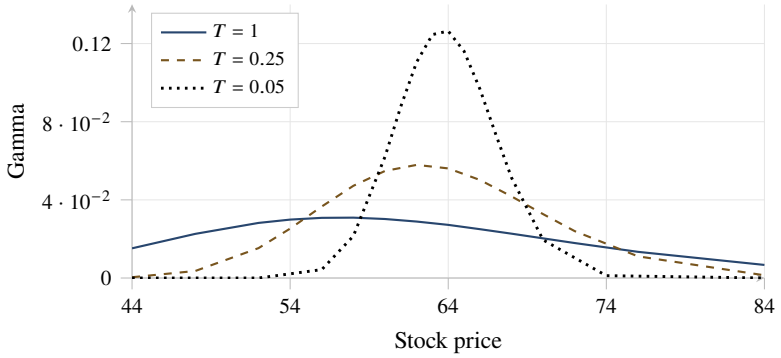


Figure 18.4: Gamma. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

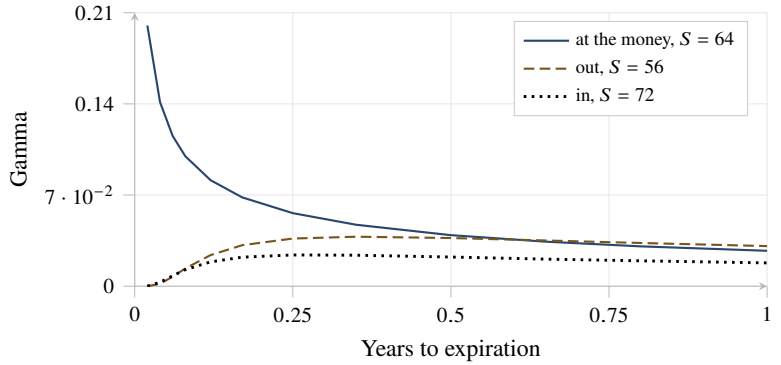


Figure 18.5: Gamma as expiration approaches. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Neutralizing gamma requires options. Delta can be hedged with the stock, because the stock's delta is 1. Gamma cannot: the stock's gamma is zero, so no quantity of it changes the book's gamma. Only another option can offset gamma, which is why dealers trade with each other and not only with clients, and why a desk short gamma across its whole book cannot reduce that exposure cheaply.

18.4 THETA, AND WHY IT IS GAMMA UNDER ANOTHER NAME

Theta, the coefficient of Δt in (18.1), is the rate at which an option loses value as time passes:

$$\Theta = \frac{\partial f}{\partial t} \tag{18.6}$$

For a European option on a non-dividend stock,

$$\Theta_{\text{call}} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2), \qquad \Theta_{\text{put}} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}N(-d_2) \tag{18.7}$$

Both share the first term, the decay of time value, and differ in the second, the interest on the strike, negative for the call and positive for the put. Theta is usually quoted per calendar day, and for the dealer's option $\Theta = -5.8757$ per year, or about $-\$0.016$ a day.

Figure 18.6 shows the shape. Set beside Figure 18.4 it is close to a mirror image (both concentrate near the money, and both sharpen as expiration nears), and the rest of this section shows that the resemblance is an identity rather than a coincidence. Theta is not a risk in the sense delta and gamma are, because the passage of time is not uncertain; what matters is its tie to gamma.

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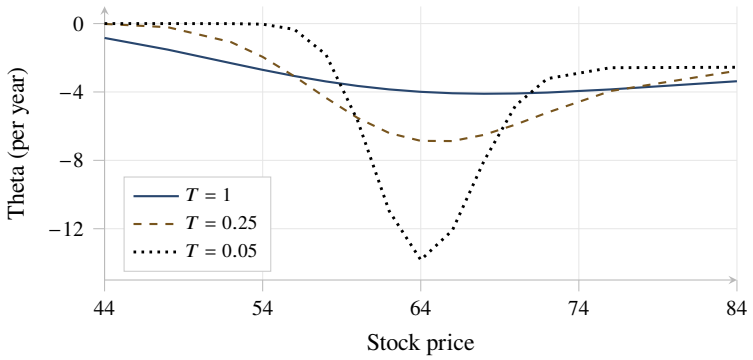


Figure 18.6: Call theta, per year. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Recall the Black–Scholes–Merton equation (15.1):

$$\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rf$$

This holds for every derivative the model prices. For a *delta-neutral* portfolio, $\Delta = 0$ and the equation reduces to

$$\boxed{\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r\Pi} \tag{18.8}$$

Theta and gamma are two sides of one quantity. A delta-neutral book long gamma normally pays theta and one short gamma normally earns it, with the trade-off fixed by the model rather than chosen by the trader. The word is *normally* because, where gamma is nearly zero, the interest term in (18.8) is all that remains, which is how a deep in-the-money put carries positive gamma and positive theta at once. Example 18.4 checks the identity on the dealer's option.

Example 18.4 Verifying the identity

For the dealer's option: $\Theta = -5.8757$, $\Delta = 0.51995$, $\Gamma = 0.048862$, $S = 63$, $\sigma = 0.22$, $r = 0.04$, $f = 3.1944$.

$$\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = -5.8757 + 0.04 \times 63 \times 0.51995 + \frac{1}{2} \times 0.0484 \times 63^2 \times 0.048862$$
$$= -5.8757 + 1.3103 + 4.6932 = 0.1278$$

and $rf = 0.04 \times 3.1944 = 0.1278$, exactly.

A trader long gamma is paid whenever the stock moves, by (18.5), and pays theta every day it does not. A trader short gamma collects theta daily and pays whenever the stock moves. Neither is a riskless profit; the model prices the exchange, and (18.8) is the price.

This also explains the standing warning of Section 12.11. A portfolio of butterflies, condors, and calendar spreads is not diversified: all are short gamma, so all collect theta and all lose together on a large move. Equation (18.8) says they are the same position.

The break-even move. That exchange has a break-even. A delta-hedged position also carries its financing, the interest on the borrowing in Table 18.1's cumulative cost column, and that is the $r\Pi$ of (18.8). Net of it, the position's carry is $\Theta - r\Pi = -\frac{1}{2}\sigma^2 S^2 \Gamma$. Setting the gamma gain of (18.5) against that carry over one trading day, $\Delta t = 1/252$ on Section 15.8's clock, gives the move at which a delta-hedged position comes out level:

$$\frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2}\sigma^2 S^2 \Gamma \, \Delta t \qquad \Longrightarrow \qquad |\Delta S| = \sigma S \sqrt{\Delta t} \tag{18.9}$$

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For the dealer's option that is $63 \times 0.22/\sqrt{252} = \$0.87$, or 1.39% of the \$63 spot. On a day the stock moves less than that the written position is ahead; on a day it moves more it is behind, and the whipsaw of Example 18.2 was a run of the second kind. Setting the gain against the option's theta alone, $|\Theta|/365$, would give \$0.81, which leaves out the financing and counts weekends as days on which the stock moves.

Gamma cancels, so the break-even is one standard deviation of the stock's daily move at the implied volatility, whatever the option. That is what it means for the option to have been fairly priced, and why a desk that has sold gamma is short realized volatility against the implied volatility it sold.

Which volatility to hedge at. Hedged at its implied volatility σ_i, a long option earns the gap between realized and implied variance day by day, weighted by gamma (Further Question 18.15), so its total depends on the path as well as on how much the stock moves. If the forecast is right, hedging at the volatility σ_r the trader expects to be realized locks in the total instead, at $f(\sigma_r) - f(\sigma_i)$, the option's value at that volatility less its value at the implied one; the position's marked value still swings until expiration (Ahmad and Wilmott, 2005). Most desks hedge at implied, the volatility the book is marked at.

18.5 VEGA

Vega measures sensitivity to volatility and is the coefficient of $\Delta\sigma$ in (18.1):

$$\mathcal{V} = \frac{\partial f}{\partial \sigma} = S_0\sqrt{T}\,N'(d_1) \tag{18.10}$$

It is positive for all purchased options, calls and puts alike, as Section 11.1 established without a model.

For the dealer's option, $\mathcal{V} = 14.769$, meaning a one-percentage-point rise in volatility raises each option by \$0.148, or \$14,769 across 100,000 options. Set that beside the \$319,438 premium: a two-point move in implied volatility changes the position's value by a tenth of what the dealer was paid to take it on. Two features matter in practice.

Vega is largest for long-dated options, scaling roughly with $\sqrt{T}$, which is the opposite of gamma. A book can therefore be short gamma and long vega at once, and desks routinely are.

Vega should not exist. The Black–Scholes–Merton model assumes σ is a known constant, so a sensitivity to changes in it is a sensitivity to something the model says cannot happen. That the market computes and hedges it anyway is the clearest admission that the model is being used as a quoting convention rather than as a description, which is Section 15.7's point, now visible in the risk report. Chapter 19 confronts the consequences.

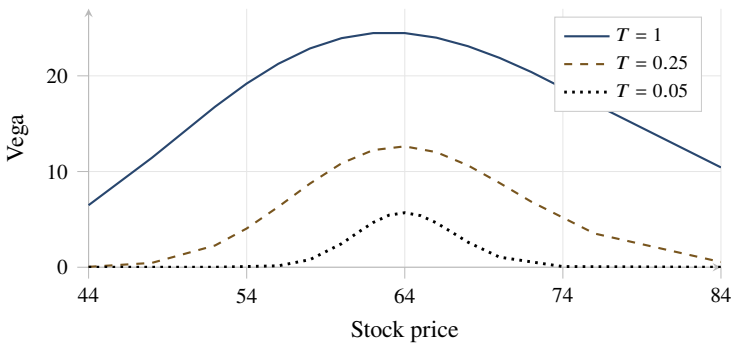


Figure 18.7: Vega. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Like gamma, vega peaks near the money (Figure 18.7); unlike gamma, it does not spike as expiration approaches but shrinks. Comparing with Figure 18.4 makes the point that governs

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how books are constructed: the short end carries the gamma and the long end carries the vega, so a desk's two largest risks live at opposite ends of the maturity curve and must be hedged with different instruments.

18.6 RHO AND THE REST

Rho measures sensitivity to the interest rate and supplies the $\rho\,\Delta r$ term that (18.1) sets aside:

$$\rho_{\text{call}} = KTe^{-rT}N(d_2), \qquad \rho_{\text{put}} = -KTe^{-rT}N(-d_2) \tag{18.11}$$

Only the discounted strike in Chapter 15's formula carries r, so rho is the time to expiration multiplied by the present value of what the holder expects to pay. For the dealer's option $\rho = 10.233$, so a one-point rise in rates is worth \$0.102 an option, or \$10,233 across the 100,000.

Set that beside vega: a one-point move in implied volatility is worth \$14,769 on the same position, the same order of magnitude. Rho is ignored on short-dated equity books not because the sensitivity is small but because the input barely moves: implied volatility travels several points in a week, policy rates in 25-basis-point steps a few times a year.

Maturity reverses that. Rho grows with T while an option's value grows roughly with $\sqrt{T}$, so the exposure per dollar of premium climbs with the term:

At-the-money call	Value	Rho	Per 100bp, % of value
18 weeks	3.739	11.29	3.0%
1 year	6.841	32.51	4.8%
5 years	18.008	147.71	8.2%
10 years	27.301	253.04	9.3%

So rho is watched wherever the term is long or the rate is the point: the ten-year employee options and warrants of Chapter 17, the convertible bonds of Chapter 25, the structured notes of Chapter 35; and the conversions, reversals and boxes of Chapter 11, which are option structures assembled so that the financing rate is the only exposure left, and which market makers use to lend and borrow. In Chapter 29's rate derivatives rho stops being a secondary Greek altogether, because there the curve is the underlying.

In currency options rho is first-order. An option on an exchange rate has two rhos, one per currency, and Chapter 16's substitution gives both:

$$\rho = KTe^{-rT}N(d_2), \qquad \rho_f = -STe^{-r_fT}N(d_1)$$

where r is the domestic interest rate, r_f the foreign one, and S the exchange rate, as in Chapter 16. They almost cancel. Adding them gives exactly $-Tc$, so a parallel move in both rates does no more than discount the option, and what survives is the exposure to the *differential*, which is what sets the forward. Take Chapter 16's euro–dollar inputs: spot 1.0850, dollar rates 4%, euro rates 2.5%, volatility 8%. A one-year euro call struck at spot gains 14.4% of its premium when dollar rates rise 100 basis points and euro rates do not; the equity option of Section 18.1 gains 3.2% on the same shift, and even a three-month currency option gains 7.5%. That is why a central bank meeting is a scheduled volatility event in the currency market in a way it is not for a single stock.

Beyond the five, desks compute higher-order and cross sensitivities as the book requires: **vanna** ($\partial^2 f/\partial S\,\partial\sigma$, how delta moves with volatility) and **volga** ($\partial^2 f/\partial\sigma^2$, how vega moves with volatility) matter wherever the smile matters, which is Chapter 19's territory and essential for the barrier options of Chapter 34. **Charm** ($\partial^2 f/\partial S\,\partial t$, how delta moves with the passage of time) is a rounding error for a monthly option and the dominant hedging flow for the same-day options of Chapter 23, where a dealer's delta can fall by two-thirds over a session in which the index does not move. Table 18.4 collects the five.

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Table 18.4: Definitions, signs for a purchased option, and what each answers.

Greek	Derivative	Sign	Question it answers
Delta Δ	$\partial f/\partial S$	+ call, − put	How much stock to hold
Gamma Γ	$\partial^2 f/\partial S^2$	+	How often to rebalance, and what a jump costs
Theta Θ	$\partial f/\partial t$	−	What waiting costs per day
Vega $\mathcal{V}$	$\partial f/\partial \sigma$	+	What a change in implied volatility costs
Rho ρ	$\partial f/\partial r$	+ call, − put	What a rate move costs

18.7 MANAGING A BOOK

A desk does not hedge options one at a time. Every Greek is a derivative, so all are additive across positions on the same underlying, and a book of thousands of trades is five numbers.

Additivity means the book has a single expansion (18.1), whose coefficients are the summed Greeks, and hedging the book is canceling its terms in order. The order is set by which instrument reaches which term. Stock carries only the first term, and only options carry the second, so gamma is cleared with options before delta is cleared with stock; Example 18.6 does it in that order. Example 18.5 first checks the expansion against a full revaluation of the book in Table 18.5.

Table 18.5: Three positions on one stock. $S = 63$, $r = 4\%$, $\sigma = 22\%$.

Position	Quantity	Delta	Gamma	Vega
Calls K=64, 18 weeks	−100,000	−51,995	−4,886	−1,476,869
Puts K=58, 18 weeks	+50,000	−10,441	+1,761	+532,343
Calls K=70, 40 weeks	−40,000	−15,435	−1,259	−845,404
Book		**−77,870**	**−4,384**	**−1,789,929**

Desks report gamma in dollars rather than shares: the change in the position's dollar delta for a 1% move in the stock, $\Gamma S^2/100$. For this book that is $-4{,}384 \times 63^2/100 = -\$174{,}000$, the stock it must buy after every 1% rise. Vega is quoted per volatility point, one hundredth of the figures in the table, and theta per calendar day.

Example 18.5 The book's expansion against a full revaluation

One day passes ($\Delta t = 1/365$), the stock falls from \$63 to \$61 ($\Delta S = -2$), and implied volatility rises from 22% to 23% ($\Delta\sigma = 0.01$), with r unchanged at 4%. Table 18.5 gives the book's delta, gamma and vega. Its theta, from (18.7), is +601,812 per year, positive because the book is net short options. The terms of (18.1) are

Term	Book Greek	Change	P&L (\$)
$\Delta\,\Delta S$	−77,870	$\Delta S = -2$	+155,740
$\frac{1}{2}\Gamma(\Delta S)^2$	−4,384	$(\Delta S)^2 = 4$	−8,768
$\Theta\,\Delta t$	+601,812	$\Delta t = 1/365$	+1,649
$\mathcal{V}\,\Delta\sigma$	−1,789,929	$\Delta\sigma = 0.01$	−17,899
Sum of the four terms			+130,722
Full revaluation			+133,001
Residual			+2,279

The full revaluation prices each of the three positions by the Black–Scholes–Merton formula at $S = 61$, $\sigma = 23\%$ and one day less to expiration. The expansion accounts for all but \$2,279

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of the $133,001, or 1.7%. The delta term carries nearly everything. The day's theta of $1,649 is, by (18.8), largely the book's payment for being short 4,384 gamma, and by (18.9) its gamma and theta break even on a move of one daily standard deviation, $0.87; the $2 move costs more than five times the day's theta.

The residual is what the expansion leaves out: terms of third order in ΔS, and cross terms. About $1,850 of it is the cross term $(\partial^2\Pi/\partial S\,\partial\sigma)\,\Delta S\,\Delta\sigma$, the change in the book's vega as the stock moves, which the separate delta and vega terms cannot see. The expansion is local, and at this size what it misses is small; Section 18.10 takes up the moves where it is not.

Example 18.6 Neutralizing a book, in the right order

Table 18.5 gives a book that is short 77,870 delta, short 4,384 gamma, and short 1.79 million vega. Hedge it with one traded option, the 40-week call struck at 64:

	Delta	Gamma	Vega
The book	−77,870	−4,384	−1,789,929
40-week call, K=64, each	0.5692	0.03232	21.711

Gamma first, because it cannot be hedged with stock. Zeroing the book's gamma requires buying

$$\frac{4{,}384}{0.03232} = 135{,}644 \text{ options}$$

Then delta, with stock. Those 135,644 options carry $135{,}644 \times 0.5692 = 77{,}209$ of delta, which takes the book from −77,870 to −661. Buying 661 shares restores neutrality.

Vega is what remains. The hedging option carried $135{,}644 \times 21.711 = 2{,}944{,}967$ of vega, so the book moves from −1,789,929 to +1,155,038. Neutralizing gamma has flipped the sign of the vega exposure and left it large.

That is the general problem. There are three exposures and, on a single underlying, only two hedging instruments: the stock and one option. Neutralizing all three requires a *second* option of a different maturity, because gamma and vega have different term structures (Figures 18.4 and 18.7). Short-dated options are the gamma instrument and long-dated options are the vega instrument, and a desk needs both. Example 18.7 builds that hedge.

Example 18.7 Clearing gamma and vega together

Example 18.6 ended gamma-neutral but $1.16 million long vega. Two options of *different* maturities clear both at once. Keep the 40-week call and add an 8-week call at the same strike:

Instrument	Delta	Gamma	Vega	Gamma per unit vega
A: K=64, 8 weeks	0.4729	0.07321	9.835	0.00744
B: K=64, 40 weeks	0.5692	0.03232	21.711	0.00149

The last column is the whole reason two maturities are enough. The short-dated option delivers five times as much gamma per unit of vega as the long-dated one, so the two are not redundant and any combination of gamma and vega can be reached.

Writing n_A and n_B for the quantities bought, neutrality is a 2×2 system:

$$0.07321\,n_A + 0.03232\,n_B = 4{,}384$$
$$9.835\,n_A + 21.711\,n_B = 1{,}789{,}929$$

with solution $n_A = 29{,}357$ and $n_B = 69{,}145$. Both are positive: the book is short both exposures, so both options are bought.

Then delta, with stock. The two positions carry $29{,}357 \times 0.4729 + 69{,}145 \times 0.5692 = 53{,}240$ of delta, taking the book from −77,870 to −24,630. Buying 24,630 shares closes it, and all three exposures are now zero.

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Gamma and vega must be solved *together*, because every option moves both; delta is solved last, because the stock moves delta alone and so cannot disturb what has just been set. A hedge built in the other order has to be redone.

It is not free. The two option positions cost $421,218 to establish and carry a combined theta of about −$1,475 per day. That is Section 18.4 arriving as a cash flow: the desk has bought gamma, and buying gamma means paying theta every day the stock fails to move. Neutrality is a position with a running cost, not a state of rest.

Books across many underlyings

Table 18.5 aggregated positions on *one* stock. A single-stock desk may hold options on hundreds of stocks, and hedging each separately would multiply Table 18.7's transaction costs by the number of underlyings.

The standard response reuses Section 3.6's device. Convert each position's delta into an equivalent index exposure using the stock's beta,

$$\Delta_{\text{index-equivalent}} = \sum_i \beta_i \, \Delta_i \, S_i \tag{18.12}$$

where Δ_i is the book's delta in stock i, S_i is its price and β_i its beta against the index. Each term is therefore a dollar exposure rather than a share count, and the sum takes the place of βV_A in Section 3.6's hedge ratio, which divides it by the value of one futures contract to give the number to sell. One trade replaces hundreds.

The hedge removes the *market* component of the book's delta and leaves the idiosyncratic component, the part of each stock's movement unexplained by the index. That residual is small on any one day and is not diversified away, because the book's positions are not random: a desk accumulates whatever clients want to sell.

The same decomposition defines a trade. A book long single-stock options and short index options is long idiosyncratic volatility and short market volatility, which is the dispersion position Section 16.2 named. A desk that hedges by (18.12) without noticing has taken that position by accident; the position pays off when correlations fall and loses when they rise, which is when everything else is losing too.

Greeks for other underlyings

Chapter 16 extended the pricing formula to indices, currencies, and futures by the substitution $S_0 \to S_0 e^{-qT}$. The Greeks follow the same route, and Table 18.6 gives the results. The simplest case is a forward contract: its delta is e^{-qT}, or 1 without a yield, and its gamma is zero, so its hedge never has to chase the stock.

Table 18.6: Call Greeks for an underlying with yield q.

Greek	Formula
Delta	$e^{-qT} N(d_1)$
Gamma	$\dfrac{e^{-qT} N'(d_1)}{S_0 \sigma \sqrt{T}}$
Theta	$-\dfrac{S_0 N'(d_1) \sigma e^{-qT}}{2\sqrt{T}} + qS_0 N(d_1) e^{-qT} - rKe^{-rT} N(d_2)$
Vega	$S_0 \sqrt{T}\, e^{-qT} N'(d_1)$

Two consequences follow. Delta is no longer bounded by 1 but by e^{-qT}: for an index option with $q = 3\%$ and $T = 2$, the factor $e^{-qT} = 0.942$ means a deep in-the-money call requires only 0.94 units of the index, not 1. Currency desks often quote the premium-adjusted delta

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of Section 16.4 instead, so a quoted delta must be read with its convention. And for a *futures* option, $q = r$ makes every Greek a derivative with respect to the futures price F: the delta is $e^{-rT}N(d_1)$, the discounting of a payoff settled on F, and the hedge is held in the futures contract itself. The factor in Section 18.9's futures hedge (18.13) is a different adjustment, needed only when a spot-based option is hedged with futures.

18.8 WHAT DESKS ACTUALLY HEDGE

A textbook can neutralize every Greek. A desk cannot, and the choices it makes are informative.

Delta is hedged, continuously and almost automatically. It is cheap, since the underlying is the most liquid instrument available, and an unhedged delta swamps every other exposure.

Gamma is managed, not neutralized. Neutralizing it requires buying options, which costs theta by (18.8). Desks therefore run a gamma position deliberately, tightening it before events and letting it drift otherwise, and set limits rather than targets.

Vega is hedged by bucket. A single vega number is nearly useless because volatility does not move in parallel: the front of the curve moves more than the back, as Chapter 19 documents. Desks compute vega by maturity bucket and hedge each, which is the same bucketing logic Section 29.10 applies to interest rate deltas.

Rho is usually ignored in equities. It dominates in rates, where Chapter 29 takes it up.

18.9 HOW OFTEN TO REBALANCE

Table 18.3 showed hedging error falling as $1/\sqrt{n}$ with no transaction costs. Introducing costs changes the problem qualitatively, because they accumulate with every trade. Table 18.7 repeats that simulation, charging 0.10% of the value of each rebalancing trade, though not of the final settlement at expiration.

Table 18.7: Cost of hedging 1 written call, with a 0.10% proportional trading cost. 20,000 paths; formula value 3.1944.

Rebalances n	Mean cost	Std. dev.	Mean + s.d.
5	3.2628	1.1924	4.4552
10	3.2926	0.8634	4.1560
20	3.3158	0.6212	3.9370
50	3.3685	0.4075	3.7760
100	3.4266	0.2976	**3.7242**
250	3.5457	0.2232	3.7689

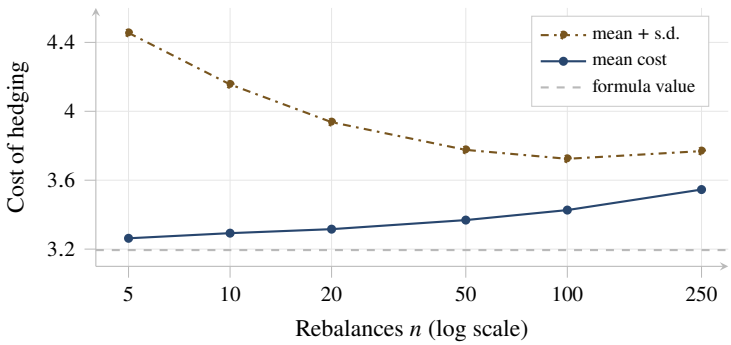


Figure 18.8: Hedging cost against rebalancing frequency, with a 0.10% trading cost.

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The two columns move in opposite directions (Figure 18.8). Trading more often reduces the *uncertainty* of the outcome (the standard deviation falls from 1.19 to 0.22) while raising its *expected* cost, from 3.26 to 3.55, as commissions accumulate. The total is minimized near $n = 100$. Three conclusions follow.

There is a finite optimum. Once costs exist, continuous rebalancing is not merely impractical but wrong: the limit is strictly worse than a finite frequency.

Hedging is not free even at the optimum. At $n = 100$ the mean cost is 3.4266 against a formula value of 3.1944, so the dealer must charge about 7.3% above the model price to break even on trading costs. That margin is one component of the bid–ask spreads of Example 10.3, and it is why illiquid underlyings have wider option markets.

A clock is the wrong trigger. Table 18.7 rebalances at fixed intervals, which trades when nothing has happened and fails to trade when a great deal has. Desks instead rebalance when delta drifts outside a band, which saves trades in quiet markets and trades more often in volatile ones. Figure 18.4 shows what the band means for the stock: where gamma is high, each dollar the stock travels moves delta further, so the stock can travel less far before the band is crossed.

Hedging with futures instead

A desk hedging an index option uses futures rather than the basket, with Section 3.6's adjustment. Because a futures price moves $e^{(r-q)(T^*-t)}$ times as far as the spot for a maturity T^*, the futures position required, in units of the underlying (dividing by the contract size gives contracts), is

$$N_{\text{futures}} = e^{-(r-q)(T^*-t)}\,\Delta_{\text{portfolio}} \tag{18.13}$$

Futures are more liquid and cheaper to trade than a basket, and they require margin rather than the full purchase price. The cost is basis risk (Section 3.3): the hedge tracks the futures price, and the option settles against the index. Where index futures trade persistently away from fair value, as China's index futures did after the 2015 trading restrictions, that basis is a running cost of a short futures hedge, charged at every roll.

18.10 BEYOND THE GREEKS

The Greeks are derivatives, so they describe behavior *near* the current state, where they are most reliable and least needed.

They fail for large moves. Equation (18.5) is a second-order approximation. Example 18.3 showed it accurate to within two cents for \$4 moves on a \$63 stock. For a 20% move it predicts a loss of \$3.88 an option either way, where repricing gives \$3.48 on a fall and \$3.07 on a rise, because gamma itself changes over so large a move. A risk report built entirely from Greeks describes a world in which nothing much happens, which is why every desk supplements them with **scenario analysis**: revalue the whole book at a grid of underlying prices and volatilities, and read the worst cell.

A typical grid moves the underlying by ±2%, ±5%, ±10%, ±20% and volatility by ±2, ±5, ±10 points, and reports the full revaluation rather than an approximation. The largest losses often sit in the corners, a big move *and* a volatility spike, where the cross term in $\Delta S\,\Delta\sigma$, which (18.1) omits, grows with the product of the two.

They assume you can trade. Every number in this chapter assumes the hedge can be adjusted at a quoted price. A gap denies it: a stock that closes at 63 and opens at 50 never traded at 56, so the delta is corrected after the loss rather than during it. Rebalancing more often does not help: Table 18.3's error falls as $1/\sqrt{n}$ only because the simulated path is continuous. On the dealer's position a 20% gap down costs \$347,821 against the \$319,438 collected, and a 30% one costs \$663,836, the "roughly twice the premium" for which Section 18.1 rejected the naked position. The hedge is also financed and consumes liquidity: Table 18.1's cumulative

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cost column is borrowing, and Sections 18.11 and 18.12 describe what happens when every desk needs the same trade at the same moment.

They assume the inputs move independently. They do not: equity volatility rises when prices fall, which is the skew of Section 15.11 restated as a dynamic. The cross-derivative vanna of Section 18.6 captures this, and a book that is delta- and vega-neutral but vanna-exposed will discover the exposure on the day both move. Hull and White (2017), estimating the correction (19.4) from index and stock option data, find a minimum-variance delta below the Black–Scholes delta for calls and puts alike: fewer shares against a call, more against a put.

They assume the model. Every Greek in Table 18.4 is a derivative *of the Black–Scholes–Merton formula*. If the formula is the wrong map from prices to values, as Section 15.11 measured, its derivatives inherit the error. Where no closed form exists, the Greeks are themselves estimated, by bumping and revaluing, by pathwise or likelihood-ratio estimators, or by algorithmic differentiation, and Sections 20.10 and 32.13 show what each costs.

The Greeks remain indispensable, but they are a local linearization of an inexact model.

18.11 CREATING OPTIONS SYNTHETICALLY

Section 18.2 described a dealer hedging a written option. Running the same machinery in reverse manufactures an option that no one sold. That is what portfolio insurance did in 1987 (Section 40.2).

To create a synthetic long put on an index portfolio with yield q, hold the portfolio and maintain a short index position worth

$$e^{-qT}\,[1 - N(d_1)] \times \text{portfolio value}$$

taken in futures through (18.13) and adjusted continuously. As the market falls, $N(d_1)$ falls and the short grows; as it rises, the short shrinks. At expiration the combination has the payoff of portfolio-plus-put.

The attraction is that no premium is paid up front; Section 12.10 explained why that matters more to a committee than it should. The cost appears instead as the accumulated trading losses of Example 18.2. Two differences from buying a put are decisive.

The cost is uncertain rather than fixed. Buying a put fixes the price at inception. Replicating one pays the realized cost, which by Table 18.7 has a standard deviation that does not vanish. If realized volatility exceeds the volatility the program assumed, replication costs more than the put would have, exactly when protection is needed most.

The replication requires a liquid market, and it consumes liquidity. Selling futures into a decline is the mechanism, so the strategy needs a market able to absorb the sales at the moment everyone running the same program is selling. Buying a put transfers this problem to the writer; replicating one keeps it.

18.12 WHEN HEDGING MOVES THE MARKET

The Greeks describe how a dealer responds to the market. In aggregate those responses *are* part of the market, and the mechanism earlier chapters met without naming is gamma.

Consider the sign. A dealer who is **short gamma**, having sold options, must buy the underlying as it rises and sell as it falls, by (18.3). That is trading *with* the move, which amplifies it; a **long-gamma** dealer does the opposite, which damps it. Five episodes in this book are now one phenomenon.

Pinning (Example 10.6). Near expiration, dealers long gamma at a strike buy below it and sell above it, holding the stock toward the strike. This is long-gamma damping, concentrated by the enormous gamma of Figure 18.4's shortest-dated curve.

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Retail call buying (Section 10.12). Retail investors buy calls, leaving dealers short gamma, and dealer hedging amplifies moves: the opposite sign, and no contradiction, since which one prevails depends on which side of the options dealers hold.

Snowball knock-ins (Sections 12.12 and 35.7). Here the sign changes with the index. Issuers hold the barrier put the note's buyer has written, so above the barrier they are long gamma and buy as the index falls, damping the decline exactly as pinning does. At the barrier that reverses discontinuously: delta jumps, part of the hedge is sold, and the bid that had been supporting the market weakens.

Portfolio insurance (Section 40.2). This is the 1987 case, and the largest. Manufacturing a long put by trading means trading with the move, selling into declines: a short-gamma dealer's flow, executed by the option's would-be buyer rather than its writer.

Inverse volatility products (Section 22.8's case study). To keep a constant inverse exposure to VIX futures, an exchange-traded product must buy those futures when VIX rises, a short-gamma hedger's flow written into a prospectus; on February 5, 2018, that buying pushed the futures higher still, and the largest such product was terminated. Here the hedge was the product itself, so no dealer's risk report showed it.

Case Study 18.1 Reading dealer gamma

Because the sign of dealer gamma determines whether hedging damps or amplifies, estimates of aggregate dealer gamma have become a widely followed market indicator.

The estimates require assumptions that deserve skepticism. Open interest is observable, but who is long and who is short is not, and the usual convention (that dealers are long the calls, bought from investors writing calls against stock they hold, and short the puts, sold to institutions buying protection) is a guess that can be badly wrong. Retail call buying since 2020 has left dealers short many of the calls the convention counts as long; Chapter 23 shows how much of the answer the assumption decides.

The mechanism is nevertheless real: dealer hedging transmits option-market flow into the underlying, in a direction that depends on a position no one observes. That is enough to explain why volatility around expirations behaves differently and why large structured-product programs leave visible marks, but not to forecast a market.

18.13 STATIC HEDGING

Everything so far has been **dynamic** hedging: hold a position, watch it drift, trade to restore it. Chapter 12 supplied the alternative, for when dynamic hedging breaks down.

Static hedging replicates a payoff with a portfolio of options bought once and held. Section 12.8 gave the recipe: any piecewise-linear payoff is a bond, a forward, and calls weighted by the change in slope at each kink. Such a hedge needs no rebalancing, incurs no gamma cost, and is indifferent to tomorrow's liquidity.

Static hedges use options, which are expensive and available only at traded strikes and maturities; dynamic hedges use the underlying, which is cheap and always available. One pays up front for certainty, the other as it goes for flexibility.

The choice is forced at discontinuities. A digital payoff is a step, so its gamma diverges near the strike at expiration. There, dynamic hedging becomes impractical, as it does near a barrier (Chapter 34); the usual answer is a static hedge, a tight call spread (Section 12.8), accepting a small mismatch for never having to trade. The snowball episodes of Sections 12.12 and 35.7 show the cost on exactly this point: issuers hedging dynamically where gamma diverged had to trade at the barrier, and all at once.

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SUMMARY

A dealer who writes options and does nothing, or who buys the underlying outright, holds a directional bet in either case. Neither is a hedge. The Greeks are the coefficients that say how a book's value responds to each of the things that move it, and hedging is the business of controlling them.

Delta, $\partial f/\partial S$, is $N(d_1)$ for a call, and holding delta shares per option written makes a book insensitive to small moves. Scaled by S/f, delta becomes the option's elasticity, its leverage, which is highest out of the money. A delta hedge buys high and sells low by construction, and the accumulated cost of doing so is the option premium: that is what the premium pays for. Against a gap, though, there is no moment at which to trade.

Gamma measures how fast delta changes. A delta-neutral book gains or loses in proportion to the square of the move, so a short-gamma position loses on moves in either direction. Gamma peaks at the money near expiration and can be neutralized only with other options. Theta is gamma under another name: for a delta-neutral portfolio the Black–Scholes–Merton equation reduces to $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma = r\Pi$, so a book long gamma pays for waiting and is paid when the market moves, and a short-gamma book is the reverse. With its financing counted, a delta hedge comes out level when the stock moves by exactly the amount its implied volatility assigns.

Vega is largest for long-dated options, the opposite of gamma, so books are commonly short gamma and long vega at once, and the very existence of vega admits that the model is a quoting convention rather than a description of the world. Rho is ignored on short-dated equity books because rates barely move, but it is first-order in currency options, where the forward is a rate differential.

Desks hedge delta continuously, manage gamma within limits rather than neutralizing it, and hedge vega bucket by bucket because volatility does not move in parallel. Hedging error falls with the number of rebalances while transaction costs rise with it, so optimal rebalancing is finite and is usually triggered by a delta band rather than a clock. A book's gamma and vega are cleared together with options of two maturities, and delta last with stock. The Greeks are local: they understate large moves and say nothing about gaps, so desks revalue the book on a grid of scenarios.

Dealer hedging feeds back into the underlying, with the sign set by gamma. Pinning, retail-driven amplification, snowball knock-ins, 1987's portfolio insurance, and the inverse volatility products of February 2018 are one mechanism seen five times, differing only in who holds which side. Manufacturing a put by trading replaces a known premium with an uncertain cost; where gamma diverges, at a digital's strike or a barrier, a static hedge in options is the answer.

FURTHER READING

Taleb's *Dynamic Hedging* is the standard practitioner account and covers what textbooks smooth over, particularly the behavior of the Greeks near barriers and expirations. On transaction costs and optimal rebalancing, the Leland (1985) adjustment is the place to start and shows explicitly how the effective volatility at which a writer must price rises with trading costs. For the feedback effects of Section 18.12, the Brady Commission's 1988 report is the primary source on the mechanism operating at scale, and Grossman (1988) analyzed what dynamic hedging does to the market it trades in, which Section 18.12's argument follows.

PRACTICE QUESTIONS

18.1 A dealer writes 50,000 calls with delta 0.62. How many shares must be held to be delta-neutral, and what happens to that number as the stock rises?

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18.2 Explain why neither a naked nor a covered position is a hedge, using the two outcomes in Section 18.1.

18.3 A delta-neutral portfolio has gamma $-4{,}000$. Estimate the profit or loss if the underlying moves by \$3, and state whether the direction of the move matters.

18.4 Using (18.8), find the theta of a delta-neutral portfolio worth \$500,000 with gamma $-2{,}000$, when $S = 80$, $\sigma = 25\%$, and $r = 4\%$.

18.5 Explain why gamma cannot be hedged with the underlying asset but delta can.

18.6 An option has vega 18. By how much does its value change if implied volatility rises from 22% to 24%? On a position of 20,000 options, what is the total?

18.7 Explain why a book can be short gamma and long vega simultaneously, and which maturities contribute each.

18.8 A trader holds a portfolio of butterflies and calendar spreads and describes it as diversified. Using (18.8), explain why it is not.

18.9 Using Table 18.7, state which rebalancing frequency minimizes the mean cost plus one standard deviation, and explain why the mean cost rises with the number of rebalances while the standard deviation falls. Explain also why the formula value of 3.1944 is not attainable at any frequency.

18.10 A pension fund holds an index portfolio and wants the payoff of portfolio-plus-put without paying a premium. State the futures position it must run and how that position changes as the market falls, and give two respects in which the result differs from buying the put outright.

18.11 A dealer who is short gamma must buy the underlying as it rises and sell as it falls. State whether that damps or amplifies the move, and explain why pinning near expiration (Example 10.6) and retail call buying (Section 10.12) push in opposite directions even though both are dealer delta-hedging.

18.12 An exchange-traded product offering inverse exposure to VIX futures must buy those futures when VIX rises. Explain why its rebalancing rule has that sign, why that rule trades the way a short-gamma hedger does, and what distinguishes this case from a dealer whose hedging incidentally moves the market.

18.13 For a European put with the same inputs as the dealer's call ($S_0 = 63$, $K = 64$, $T = 18/52$, $r = 4\%$, $\sigma = 22\%$), find its value by put–call parity, then its delta and the elasticity Ω of Example 18.1. Interpret the sign of each, and compare the put's leverage with the call's.

FURTHER QUESTIONS

18.14 Section 18.9 shows hedging error falling as $1/\sqrt{n}$ while expected transaction costs rise, roughly as $\sqrt{n}$ for a proportional cost. Write down the total expected cost as a function of n with a proportional trading cost k, find the n that minimizes it, and describe how the optimum moves with volatility and with k.

18.15 Example 18.4 verifies the Black–Scholes–Merton equation from which (18.8) follows. Show that a trader who is long gamma and delta-hedges continuously earns the gamma-weighted difference between realized and implied variance, $\int \tfrac{1}{2}\Gamma S^2(\sigma^2_{\text{realized}} - \sigma^2_{\text{implied}})\,dt$ over the option's life, and explain why the weighting matters to someone who buys options because they expect a specific event.

18.16 Section 18.12 argues that dealer hedging amplifies moves when dealers are short gamma and damps them when long. Construct the conditions under which the aggregate position

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flips sign during a single trading day, and explain why this makes gamma-based indicators unreliable precisely when they are most consulted.

18.17 A desk is short a large quantity of short-dated at-the-money options into an earnings announcement. Using Figures 18.2 and 18.4, describe what happens to the book's delta and gamma across the announcement, and state what the desk should have done beforehand and why it might not have.

18.18 Vega measures sensitivity to a parameter the model assumes constant. Construct the argument that computing vega is therefore incoherent, then the argument that it is nevertheless the right thing to do, and state which position a risk manager should take.

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